

Machine Learning Methods for Neural Data Analysis

Expectation-Maximization (EM)

Final project

- **Friday, May 16:** Project proposal due (1pg).
 - Groups (2 people)
 - Project Goal / Key Idea
 - Dataset(s)
- **Weds, May 21:** Lab 7 (last lab) due.
- **Monday, June 9:** Final report due.

Types of final projects

- The final project is an opportunity to **apply what you've learned** to a **problem of interest** to you. If you're a PhD student, we hope it will advance your research.
- For example, you could:
 - **Propose and implement an extension** to an existing method to address its limitations.
 - **Implement a related method** from a recent research paper and reproduce its results.
 - **Adapt a method** from class to study a scientific question of interest on a new dataset.
- Your project must involve some modeling and/or algorithmic innovations, more than just applying code from lab to a new dataset.

Datasets

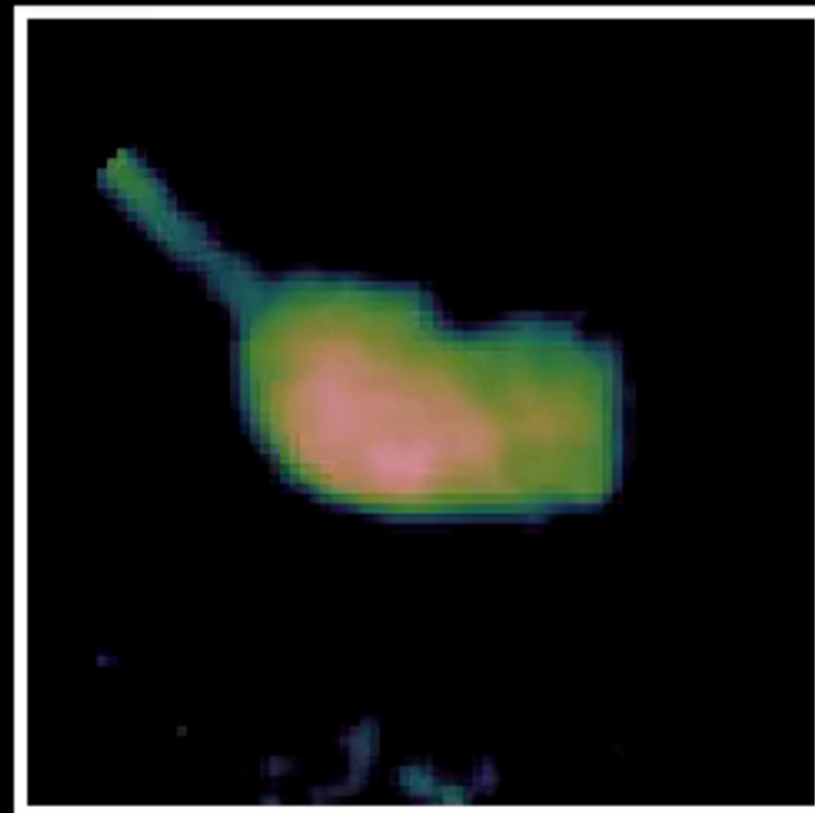
- Most recent papers in Nature, Science, Cell, etc., have publicly available data.
- If you are a graduate student and have access to a relevant dataset from your research, that should be fine too! (But talk to me first about EEG/MEG/fMRI data since this course didn't cover those modalities.)
- NeuroMatch Academy has curated some **large-scale neural and behavioral datasets**:
 - <https://compneuro.neuromatch.io/projects/neurons/README.html>
 - <https://compneuro.neuromatch.io/projects/behavior/README.html>
- NeuroMatch also has many **project templates** that could spark ideas:
 - https://compneuro.neuromatch.io/projects/docs/project_templates.html

Rough Rubric for Final Projects

- A+: Outstanding work. (Anticipating top ~10%.) Ambitious undertaking with challenging implementation of a new idea, a well-motivated question and choice of dataset. Rigorous, quantitative evaluation. Clear and detailed report.
- A: Excellent work. (~60-90th percentile.) Typically characterized by: Strong proposal that required considerable work and/or showed great creativity. Rigorous, quantitative evaluation. Clear and detailed report.
- B: Good work. (~30-60th percentile.) Typically characterized by: An interesting question, but a smaller leap from the labs. Comparisons may be more qualitative in nature. Reports may be less polished.
- C: Fair work. (~10-30th percentile.) Typically characterized by: Relatively small or extension of lab assignments. Comparisons often more qualitative in nature. Reports left much room for improvement.
- D: Bottom (~10%.) This category tends to be more variable, but is typically characterized by: Nearly a direct application of lab work. Qualitative or non-existent comparisons. Very short or sloppy report.

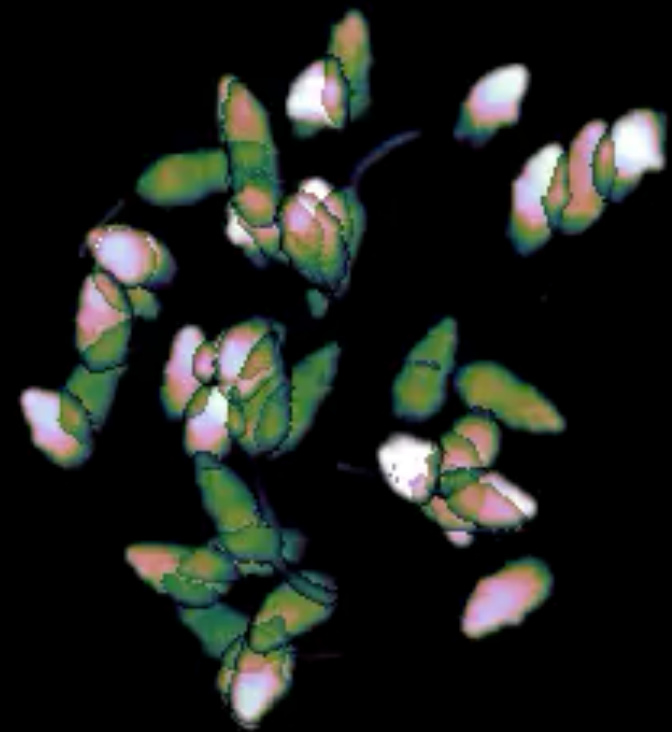
Motivating Example: summarizing videos with behavioral states

Frame 0

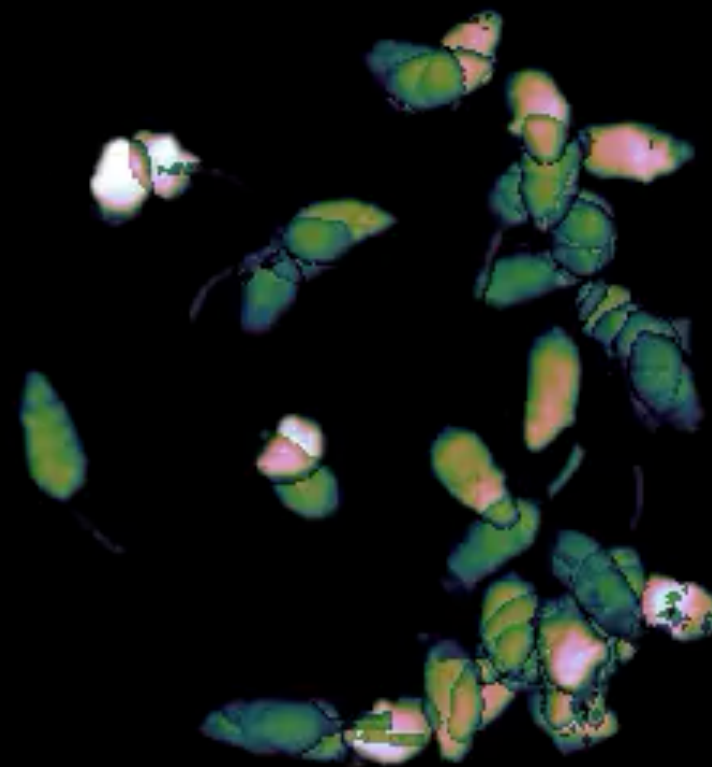


Motivating Example: summarizing videos with behavioral states

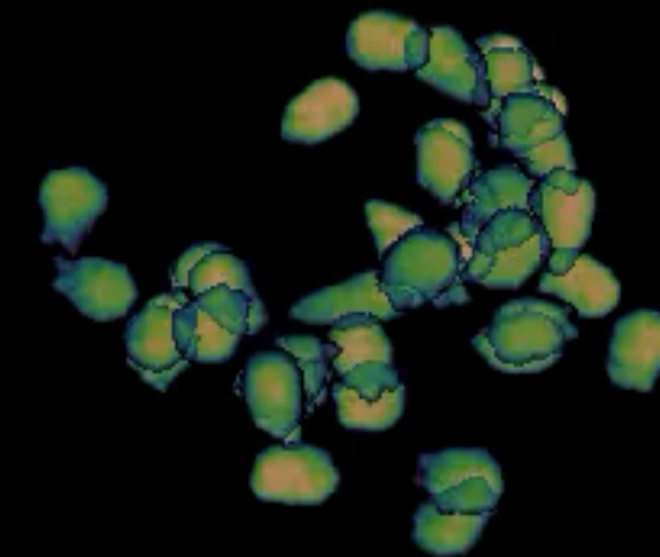
Rear down



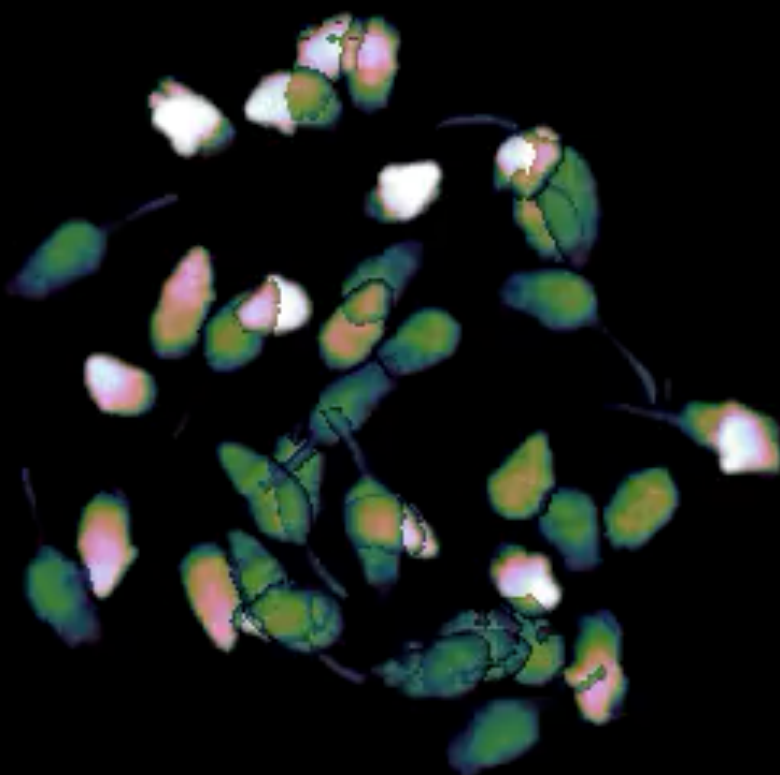
Walk forward



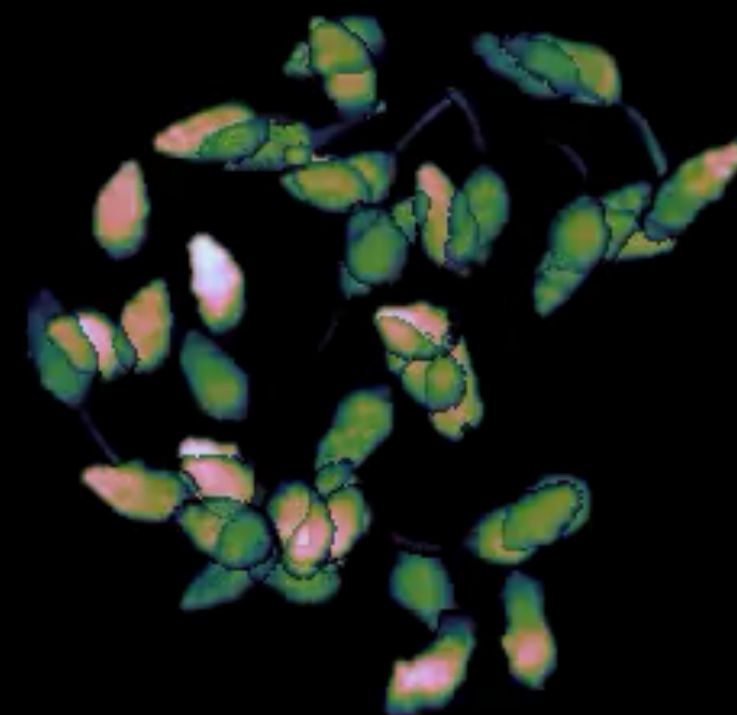
Grooming



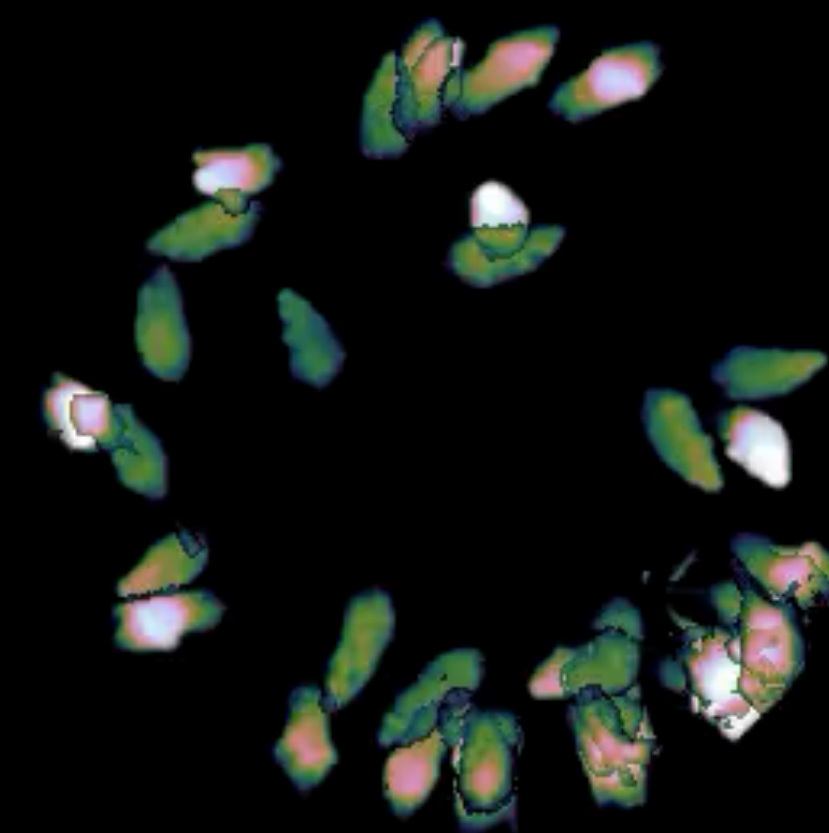
Scrunch



Rear up

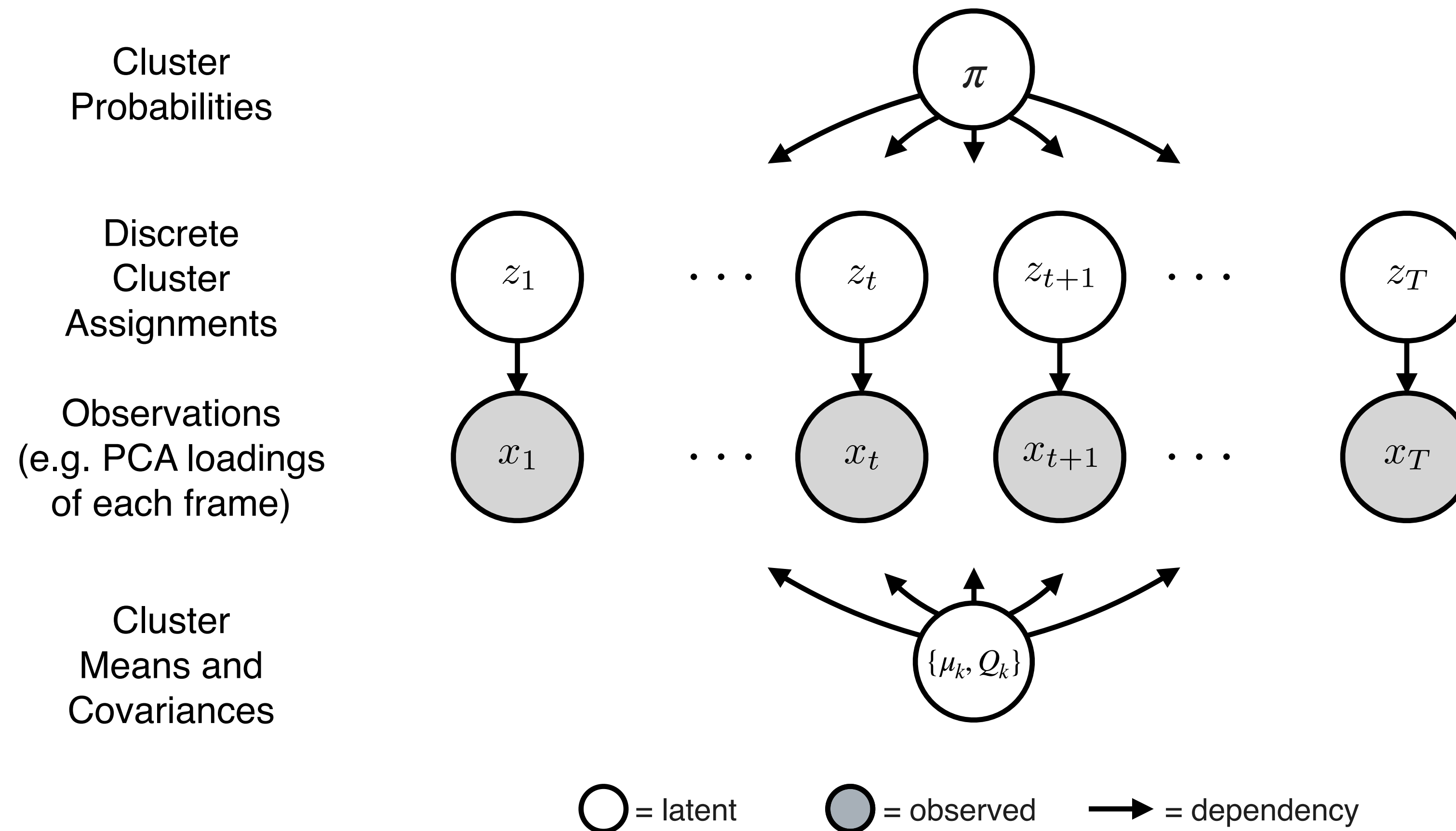


Jump



What We've Seen: The Gaussian Mixture Model

Graphical Model



What We've Seen: The Gaussian Mixture Model

Joint probability

Cluster assignments are independent across time

$$\begin{aligned} z_t &\sim \text{Cat}(\pi), & \text{for } t = 1, \dots, T \\ x_t \mid z_t &\sim \mathcal{N}(\mu_{z_t}, Q_{z_t}) & \text{for } t = 1, \dots, T \end{aligned}$$

Its parameters are $\Theta = \pi, \{\mu_k, Q_k\}_{k=1}^K$ where π is the prior distribution over clusters and (μ_k, Q_k) are the cluster means and covariances, respectively.

Under this model, the **log joint probability** factors as

$$\begin{aligned} \log p(x, z, \Theta) &= \sum_{t=1}^T \log p(z_t) + \log p(x_t \mid z_t) \\ &= \sum_{t=1}^T \log \text{Cat}(z_t \mid \pi) + \log \mathcal{N}(x_t \mid \mu_{z_t}, Q_{z_t}) \end{aligned}$$

Fitting a Gaussian Mixture Model

Coordinate Ascent

Find the cluster assignments $z_t^\star$ and parameters $\Theta^\star$ that maximize the (log) joint prob:

1. Update the cluster assignments for all time steps,

$$\begin{aligned} z_t &\leftarrow \arg \max_k \log \text{Cat}(k \mid \pi) + \log \mathcal{N}(x_t \mid \mu_k, Q_k) \\ &= \arg \max_k \log \pi_k - \frac{D}{2} \log |Q_k| - \frac{1}{2} (x_t - \mu_k)^\top Q_k^{-1} (x_t - \mu_k) \end{aligned}$$

Note that all clusters can be updated independently!

2. Update the parameters for each cluster based on the assignments. For example,

$$\begin{aligned} \mu_k &\leftarrow \arg \max_{\mu} \sum_{t: z_t=k} \log \mathcal{N}(x_t \mid \mu, Q_k) \\ &= \frac{1}{T_k} \sum_{t: z_t=k} x_t \end{aligned}$$

where $T_k = \sum_t \mathbb{I}[z_t = k]$ is the number of data points assigned to cluster k .

Fitting a Gaussian Mixture Model

K-Means

Consider the special case where $\pi = \left(\frac{1}{K}, \dots, \frac{1}{K}\right)$ and $Q_k = I$. Then coordinate ascent is equivalent to the **K-Means algorithm**:

1. Update the cluster assignments for all time steps,

$$\begin{aligned} z_t &\leftarrow \arg \max_k -\frac{1}{2}(x_t - \mu_k)^\top (x_t - \mu_k) \\ &= \arg \min_k \|x_t - \mu_k\|_2^2 \end{aligned}$$

Assign each data point to the nearest cluster.

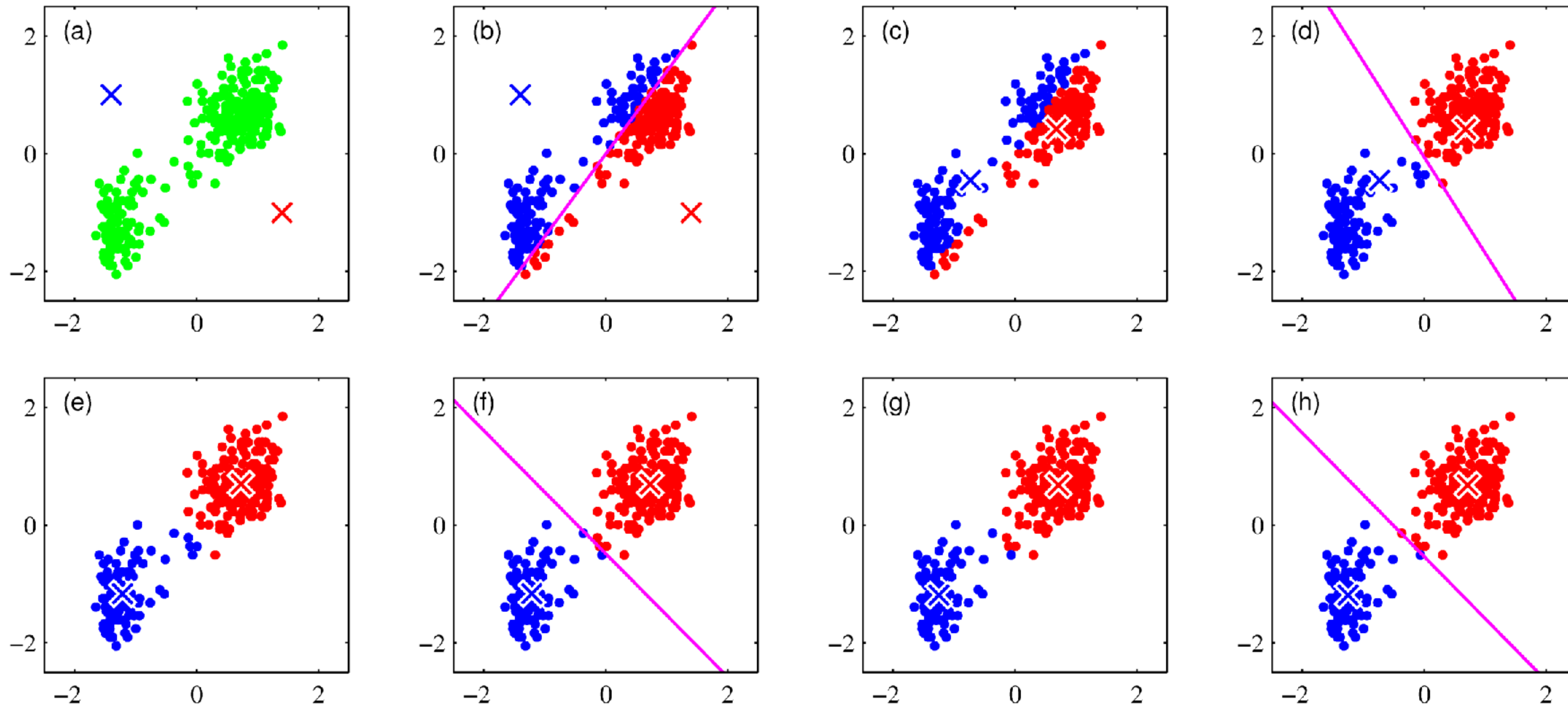
2. Update the parameters for each cluster based on the assignments. For example,

$$\begin{aligned} \mu_k &\leftarrow \arg \max_{\mu} \sum_{t: z_t=k} \log \mathcal{N}(x_t \mid \mu, Q_k) \\ &= \frac{1}{T_k} \sum_{t: z_t=k} x_t \end{aligned}$$

Set each cluster centroid to the mean of the assigned data points.

Fitting a Gaussian Mixture Model

K-Means in Pictures (Bishop, Ch 9.1)



Fitting a Gaussian Mixture Model

Expectation-Maximization (EM)

Does it really make sense to use **hard assignments** of data points to the nearest cluster? What about points on the boundary? Consider the following alternative:

1. Compute the **responsibilities**

$$q_{t,k} \triangleq q(z_t = k) \triangleq \Pr(z_t = k \mid x_t, \theta) \propto \pi_k \mathcal{N}(x_t \mid \mu_k, Q_k)$$

2. Update the parameters for each cluster based on the responsibilities. For example,

$$\mu_k \leftarrow \frac{1}{T_k} \sum_t q_{t,k} x_t \quad Q_k \leftarrow \frac{1}{T_k} \sum_t q_{t,k} (x_t - \mu_k)(x_t - \mu_k)^\top$$

where $T_k = \sum_t q_{t,k}$ is the *expected* number of data points assigned to cluster k .

Fitting a Gaussian Mixture Model

Expectation-Maximization (EM)

Does it really make sense to use **hard assignments** of data points to the nearest cluster? What about points on the boundary? Consider the following alternative:

1. Compute the **responsibilities**

$$q(z_t = k) \triangleq \Pr(z_t = k \mid x_t, \theta) \propto \pi_k \mathcal{N}(x_t \mid \mu_k, Q_k)$$

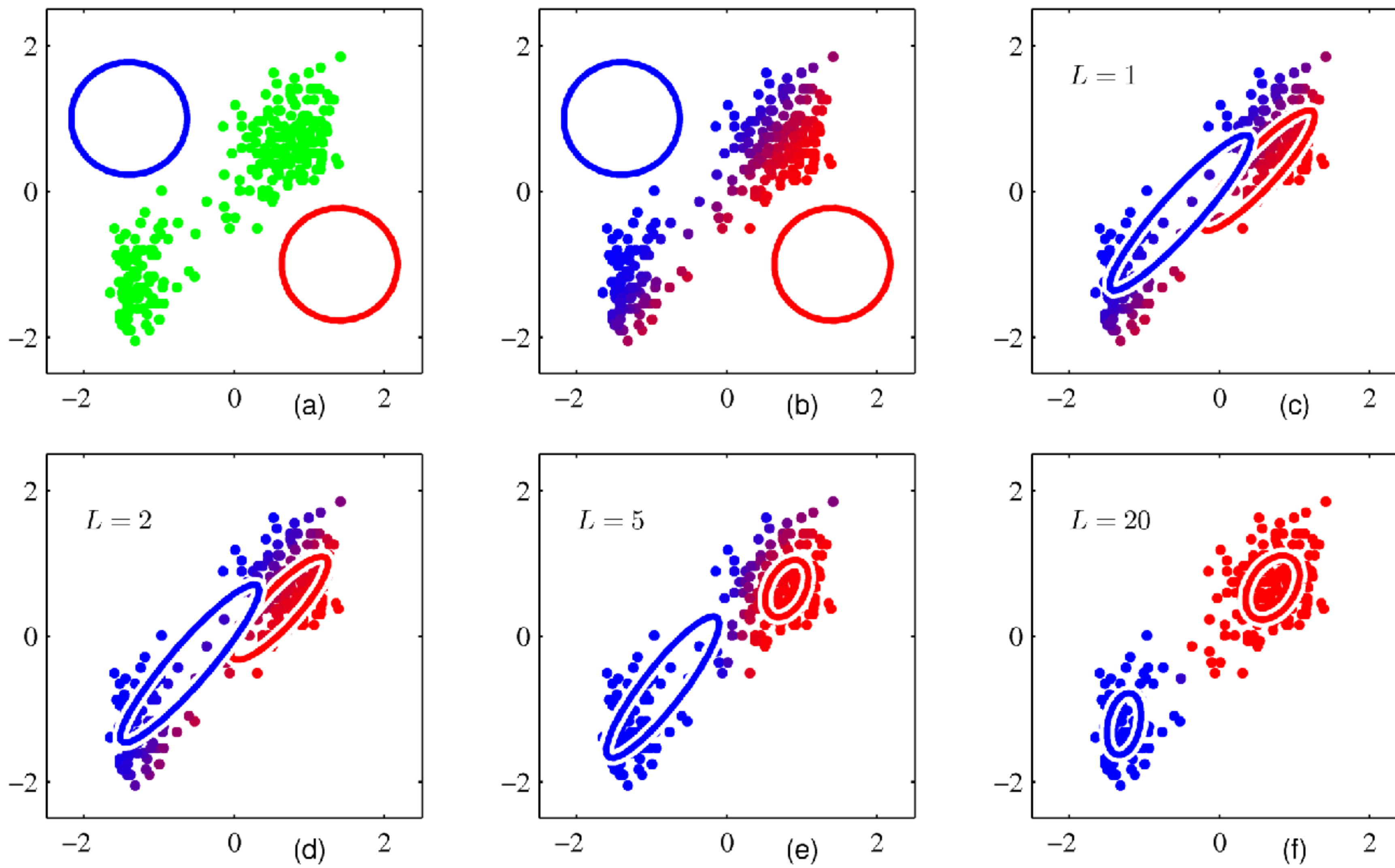
2. Update the parameters for each cluster based on the responsibilities. More generally,

$$\theta \leftarrow \arg \max_{\theta} \mathbb{E}_{q(z)} [\log p(x \mid z; \theta)]$$

This is called the **expectation-maximization (EM) algorithm**.

Fitting a Gaussian Mixture Model

EM for the GMM in Pictures (Bishop, Ch 9.1)



Another view of the EM algorithm

EM finds a local maximum of the log marginal likelihood

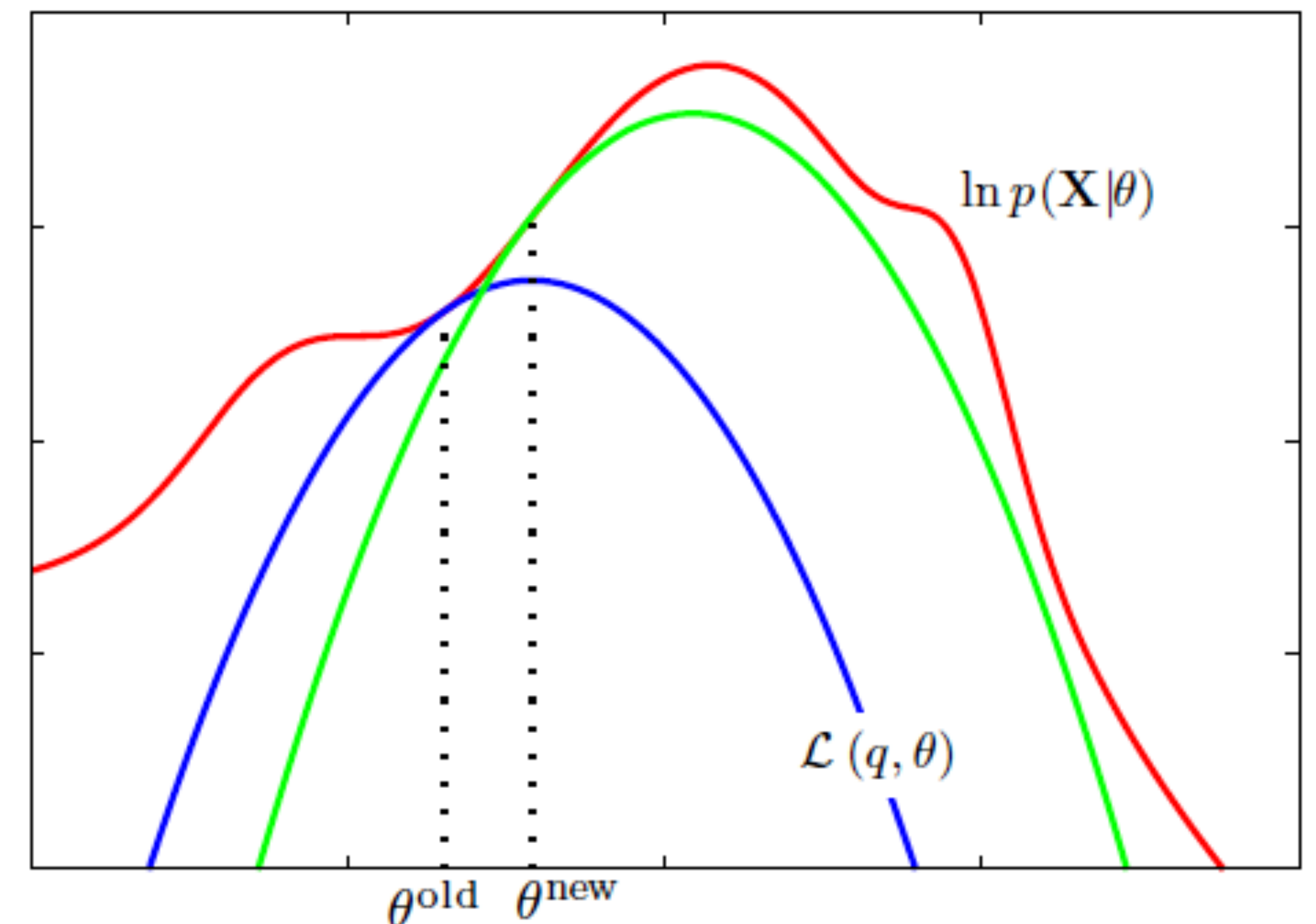
- **M-step:** Maximize the expected log probability

$$\Theta \leftarrow \arg \max_{\Theta} \mathbb{E}_{q(z)} [\log p(x, z; \Theta)]$$

- **E-step:** Update the posterior over latent variables (*these are the responsibilities*)

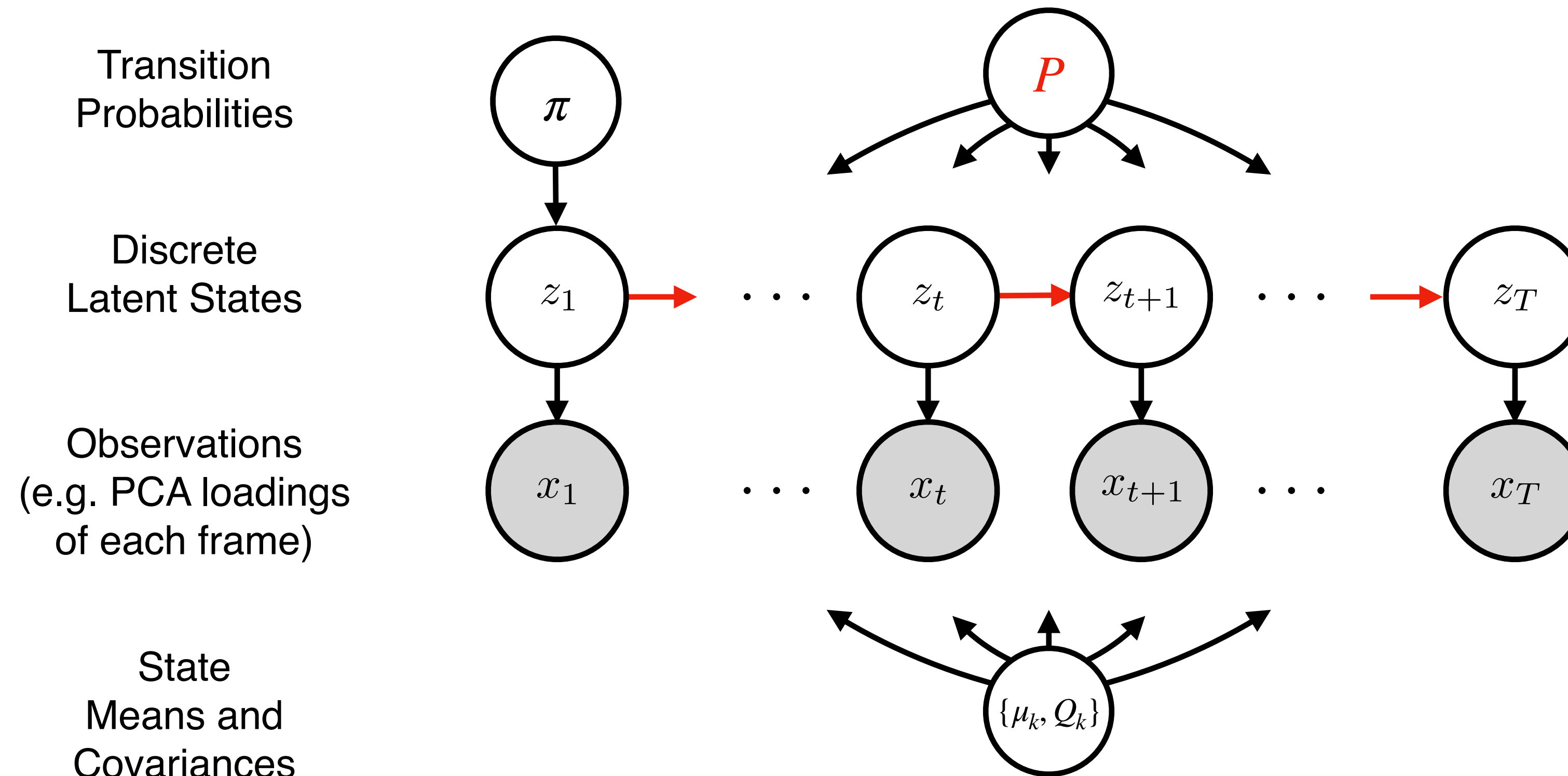
$$q(z) \leftarrow p(z \mid x, \Theta)$$

- EM converges to **local maxima** of the log marginal likelihood, $\log p(x; \Theta)$



The Gaussian HMM

Graphical Model



$\bigcirc$ = latent $\bullet$ = observed $\longrightarrow$ = dependency

The Gaussian HMM

A Gaussian HMM is just a Gaussian mixture model but where cluster assignments are linked across time!

$$\begin{aligned} z_1 &\sim \text{Cat}(\pi), \\ z_t \mid z_{t-1} &\sim \text{Cat}(P_{z_{t-1}}), \quad \text{for } t = 2, \dots, T. \\ x_t \mid z_t &\sim \mathcal{N}(\mu_{z_t}, Q_{z_t}) \quad \text{for } t = 1, \dots, T \end{aligned}$$

Its parameters are $\Theta = \pi, P, \{\mu_k, Q_k\}_{k=1}^K$ where $P \in [0,1]^{K \times K}$ is a row-stochastic **transition matrix**.

Under this model, the **joint probability** factors as

$$p(x, z, \Theta) = p(z_1) \prod_{t=1}^{T-1} p(z_{t+1} \mid z_t) \prod_{t=1}^T p(x_t \mid z_t)$$

EM for the Gaussian HMM

The M-step

In the M-step we set,

$$\Theta = \arg \max_{\Theta} \mathbb{E}_{q(z)}[\log p(x, z, \Theta)]$$

As a function of the mean μ_k and variance Q_k for cluster k , this objective is,

$$\mathbb{E}_{q(z)}[\log p(x, z, \Theta)] = \mathbb{E}_{q(z)} \left[\sum_{t=1}^T \log \mathcal{N}(x_t \mid \mu_{z_t}, Q_{z_t}) + \log \text{Cat}(z_t \mid \pi) \right]$$

EM for the Gaussian HMM

The M-step

In the M-step we set,

$$\Theta = \arg \max_{\Theta} \mathbb{E}_{q(z)}[\log p(x, z, \Theta)]$$

As a function of the mean μ_k and variance Q_k for cluster k , this objective is,

$$\begin{aligned} \mathbb{E}_{q(z)}[\log p(x, z, \Theta)] &= \mathbb{E}_{q(z)} \left[\sum_{t=1}^T \log \mathcal{N}(x_t \mid \mu_{z_t}, Q_{z_t}) + \log \text{Cat}(z_t \mid \pi) \right] \\ &= \mathbb{E}_{q(z)} \left[\sum_{t=1}^T \sum_{j=1}^K \mathbb{I}[z_t = j] \log \mathcal{N}(x_t \mid \mu_j, Q_j) \right] + \text{const} \end{aligned}$$

EM for the Gaussian HMM

The M-step

In the M-step we set,

$$\Theta = \arg \max_{\Theta} \mathbb{E}_{q(z)}[\log p(x, z, \Theta)]$$

As a function of the mean μ_k and variance Q_k for cluster k , this objective is,

$$\begin{aligned} \mathbb{E}_{q(z)}[\log p(x, z, \Theta)] &= \mathbb{E}_{q(z)} \left[\sum_{t=1}^T \log \mathcal{N}(x_t \mid \mu_{z_t}, Q_{z_t}) + \log \text{Cat}(z_t \mid \pi) \right] \\ &= \mathbb{E}_{q(z)} \left[\sum_{t=1}^T \sum_{j=1}^K \mathbb{I}[z_t = j] \log \mathcal{N}(x_t \mid \mu_j, Q_j) \right] + \text{const} \\ &= \sum_{t=1}^T \mathbb{E}_{q(z)}[\mathbb{I}[z_t = k]] \left(-\frac{1}{2} \log |Q_k| - \frac{1}{2} (x_t - \mu_k)^\top Q_k^{-1} (x_t - \mu_k) \right) + \text{const} \end{aligned}$$

EM for the Gaussian HMM

The M-step

In the M-step we set,

$$\Theta = \arg \max_{\Theta} \mathbb{E}_{q(z)}[\log p(x, z, \Theta)]$$

As a function of the mean μ_k and variance Q_k for cluster k , this objective is,

$$\begin{aligned} \mathbb{E}_{q(z)}[\log p(x, z, \Theta)] &= \mathbb{E}_{q(z)} \left[\sum_{t=1}^T \log \mathcal{N}(x_t \mid \mu_{z_t}, Q_{z_t}) + \log \text{Cat}(z_t \mid \pi) \right] \\ &= \mathbb{E}_{q(z)} \left[\sum_{t=1}^T \sum_{j=1}^K \mathbb{I}[z_t = j] \log \mathcal{N}(x_t \mid \mu_j, Q_j) \right] + \text{const} \\ &= \sum_{t=1}^T \mathbb{E}_{q(z)}[\mathbb{I}[z_t = k]] \left(-\frac{1}{2} \log |Q_k| - \frac{1}{2} (x_t - \mu_k)^\top Q_k^{-1} (x_t - \mu_k) \right) + \text{const} \\ &= \sum_{t=1}^T q_{t,k} \left(-\frac{1}{2} \log |Q_k| - \frac{1}{2} (x_t - \mu_k)^\top Q_k^{-1} (x_t - \mu_k) \right) + \text{const} \end{aligned}$$

EM for the Gaussian HMM

The M-step

Taking derivatives and setting to zero yields the following updates,

$$T_k = \sum_{t=1}^T q_{t,k}$$

$$\mu_k = \frac{1}{T_k} \sum_{t=1}^T q_{t,k} x_t$$

$$Q_k = \frac{1}{T_k} \sum_{t=1}^T q_{t,k} (x_t - \mu_k)(x_t - \mu_k)^\top$$

Note: we only need the **posterior marginal probabilities** $q_{t,k} = \Pr(z_t = k \mid x_{1:T}; \Theta)$!

EM for the Gaussian HMM

The posterior is a little trickier than in the Gaussian mixture model

- **E-step:** Update the posterior over latent variables,

$$q(z) \leftarrow p(z \mid x, \Theta) \propto p(x, z, \Theta) = p(z_1) \prod_{t=1}^{T-1} p(z_{t+1} \mid z_t) \prod_{t=1}^T p(x_t \mid z_t)$$

- The normalized posterior no longer has a simple **closed form!**
- However, we can still **efficiently compute** the **marginal probabilities** for the **M-step**.

EM for the Gaussian HMM

Computing posterior marginals

- Consider the marginal probability of state k at time t :

$$\begin{aligned} q_{t,k} &\triangleq q(z_t = k) = \sum_{z_1=1}^K \cdots \sum_{z_{t-1}=1}^K \sum_{z_{t+1}=1}^K \cdots \sum_{z_T=1}^K q(z, \dots, z_{t-1}, z_t = k, z_{t+1}, \dots, z_T) \\ &\propto \left[\sum_{z_1=1}^K \cdots \sum_{z_{t-1}=1}^K p(z_1) \prod_{s=1}^{t-1} p(x_s | z_s) p(z_{s+1} | z_s) \right] \times \left[p(x_t | z_t) \right] \\ &\quad \times \left[\sum_{z_{t+1}=1}^K \cdots \sum_{z_T=1}^K \prod_{u=t+1}^T p(z_u | z_{u-1}) p(x_u | z_u) \right] \\ &\triangleq \alpha_t(z_t) \times p(x_t | z_t) \times \beta_t(z_t) \end{aligned}$$

The M-step with sufficient statistics

EM for the Gaussian HMM

Sufficient statistics

$$\mathbb{E}_{q(z)}[\log p(x, z, \Theta)] = \sum_{t=1}^T q_{t,k} \left[-\frac{1}{2} \log |Q_k| - \frac{1}{2} (x_t - \mu_k)^\top Q_k^{-1} (x_t - \mu_k) \right] + c$$

EM for the Gaussian HMM

Sufficient statistics

$$\begin{aligned}\mathbb{E}_{q(z)}[\log p(x, z, \Theta)] &= \sum_{t=1}^T q_{t,k} \left[-\frac{1}{2} \log |Q_k| - \frac{1}{2} (x_t - \mu_k)^\top Q_k^{-1} (x_t - \mu_k) \right] + c \\ &= \sum_{t=1}^T q_{t,k} \left[-\frac{1}{2} \log |Q_k| - \frac{1}{2} x_t^\top Q_k^{-1} x_t + \mu_k^\top Q_k^{-1} x_t - \frac{1}{2} \mu_k^\top Q_k^{-1} \mu_k \right] + c\end{aligned}$$

EM for the Gaussian HMM

Sufficient statistics

$$\begin{aligned}\mathbb{E}_{q(z)}[\log p(x, z, \Theta)] &= \sum_{t=1}^T q_{t,k} \left[-\frac{1}{2} \log |Q_k| - \frac{1}{2} (x_t - \mu_k)^\top Q_k^{-1} (x_t - \mu_k) \right] + c \\ &= \sum_{t=1}^T q_{t,k} \left[-\frac{1}{2} \log |Q_k| - \frac{1}{2} x_t^\top Q_k^{-1} x_t + \mu_k^\top Q_k^{-1} x_t - \frac{1}{2} \mu_k^\top Q_k^{-1} \mu_k \right] + c \\ &= \sum_{t=1}^T q_{t,k} \left[\left\langle -\frac{1}{2} \log |Q_k|, 1 \right\rangle + \left\langle -\frac{1}{2} Q_k^{-1}, x_t x_t^\top \right\rangle + \left\langle \mu_k^\top Q_k^{-1}, x_t \right\rangle + \left\langle -\frac{1}{2} \mu_k^\top Q_k^{-1} \mu_k, 1 \right\rangle \right] + c\end{aligned}$$

EM for the Gaussian HMM

Sufficient statistics

$$\begin{aligned}\mathbb{E}_{q(z)}[\log p(x, z, \Theta)] &= \sum_{t=1}^T q_{t,k} \left[-\frac{1}{2} \log |Q_k| - \frac{1}{2} (x_t - \mu_k)^\top Q_k^{-1} (x_t - \mu_k) \right] + c \\ &= \sum_{t=1}^T q_{t,k} \left[-\frac{1}{2} \log |Q_k| - \frac{1}{2} x_t^\top Q_k^{-1} x_t + \mu_k^\top Q_k^{-1} x_t - \frac{1}{2} \mu_k^\top Q_k^{-1} \mu_k \right] + c \\ &= \sum_{t=1}^T q_{t,k} \left[\left\langle -\frac{1}{2} \log |Q_k|, 1 \right\rangle + \left\langle -\frac{1}{2} Q_k^{-1}, x_t x_t^\top \right\rangle + \left\langle \mu_k^\top Q_k^{-1}, x_t \right\rangle + \left\langle -\frac{1}{2} \mu_k^\top Q_k^{-1} \mu_k, 1 \right\rangle \right] + c \\ &= \left\langle -\frac{1}{2} \log |Q_k|, N_k \right\rangle + \left\langle -\frac{1}{2} Q_k^{-1}, \bar{\psi}_{k,1} \right\rangle + \left\langle \mu_k^\top Q_k^{-1}, \bar{\psi}_{k,2} \right\rangle + \left\langle -\frac{1}{2} \mu_k^\top Q_k^{-1} \mu_k, \bar{\psi}_{k,3} \right\rangle + c\end{aligned}$$

where

$$T_k = \sum_{t=1}^T q_{t,k} \quad \mathbf{t}_{k,1} = \sum_{t=1}^T q_{t,k} x_t \quad \mathbf{t}_{k,2} = \sum_{t=1}^T q_{t,k} x_t x_t^\top$$

are the **weighted sums of sufficient statistics**.

EM for the Gaussian HMM

Solving for the optimal Gaussian parameters

The objective we're trying to maximize is,

$$\mathcal{L}(q, \theta) = \left\langle -\frac{1}{2} \log |Q_k| - \frac{1}{2} \mu_k^\top Q_k^{-1} \mu_k, T_k \right\rangle + \left\langle \mu_k^\top Q_k^{-1}, \mathbf{t}_{k,1} \right\rangle + \left\langle -\frac{1}{2} Q_k^{-1}, \mathbf{t}_{k,2} \right\rangle + c$$

Taking the partial derivative wrt b_k and setting equal to zero,

$$\frac{\partial}{\partial \mu_k} \mathcal{L}(q, \theta) = Q_k^{-1} \mathbf{t}_{k,1} - Q_k^{-1} \mu_k T_k = 0$$

$$\implies \mu_k^\star = \frac{\mathbf{t}_{k,1}}{T_k} = \frac{1}{T_k} \sum_{t=1}^T q(z_t = k) x_t$$

EM for the Gaussian HMM

Solving for the optimal Gaussian parameters

Plug in the optimum

$$\begin{aligned}\mathcal{L}(q, \theta) &= \left\langle -\frac{1}{2} \log |Q_k| - \frac{1}{2} \frac{\mathbf{t}_{k,1}^\top}{T_k} Q_k^{-1} \frac{\mathbf{t}_{k,1}}{T_k}, T_k \right\rangle + \left\langle -\frac{1}{2} Q_k^{-1}, \mathbf{t}_{k,2} \right\rangle + \left\langle \frac{\mathbf{t}_{k,1}^\top}{T_k} Q_k^{-1}, \mathbf{t}_{k,1} \right\rangle + c \\ &= \left\langle -\frac{1}{2} \log |Q_k|, T_k \right\rangle + \left\langle -\frac{1}{2} Q_k^{-1}, \mathbf{t}_{k,2} - \frac{\mathbf{t}_{k,1} \mathbf{t}_{k,1}^\top}{T_k} \right\rangle + c\end{aligned}$$

EM for the Gaussian HMM

Solving for the optimal Gaussian parameters

Let $\Lambda_k = Q_k^{-1}$,

$$\mathcal{L}(q, \theta) = \left\langle \frac{1}{2} \log |\Lambda_k|, T_k \right\rangle + \left\langle -\frac{1}{2} \Lambda_k, \mathbf{t}_{k,2} - \frac{\mathbf{t}_{k,1} \mathbf{t}_{k,1}^\top}{T_k} \right\rangle + c$$

Taking the partial derivative wrt Λ_k and setting equal to zero,

$$\begin{aligned} \frac{\partial}{\partial \Lambda_k} \mathcal{L}(q, \theta) &= \frac{T_k}{2} \Lambda_k^{-1} - \frac{1}{2} \left(\mathbf{t}_{k,2} - \frac{\mathbf{t}_{k,1} \mathbf{t}_{k,1}^\top}{T_k} \right) = 0 \\ \Rightarrow (\Lambda_k^{-1})^\star &= Q_k^\star = \frac{1}{T_k} \left(\mathbf{t}_{k,2} - \frac{\mathbf{t}_{k,1} \mathbf{t}_{k,1}^\top}{T_k} \right) \end{aligned}$$

EM for the Gaussian HMM

In summary...

- **E-step:** Compute the posterior probabilities:

$$q_{t,k} \leftarrow p(z_t = k \mid x_{1:T}, \Theta)$$

via the **forward-backward algorithm**.

Compute **weighted sums of sufficient statistics**:

$$T_k = \sum_{t=1}^T q_{t,k} \quad \mathbf{t}_{k,1} = \sum_{t=1}^T q_{t,k} x_t \quad \mathbf{t}_{k,2} = \sum_{t=1}^T q_{t,k} x_t x_t^\top$$

- **M-step:** Update the parameters.

$$\mu_k \leftarrow \frac{\mathbf{t}_{k,1}}{T_k} \quad Q_k \leftarrow \frac{1}{T_k} \left(\mathbf{t}_{k,2} - \frac{\mathbf{t}_{k,1} \mathbf{t}_{k,1}^\top}{T_k} \right)$$

- *Note: The updates are equivalent if we use **normalized sufficient statistics**, each divided by T .*

Conclusion

- Hidden Markov models (HMMs) are just mixture models with dependencies across time.
- The EM algorithm is nearly the same as for mixture models, but we use the **forward-backward algorithm** to compute posterior marginal probabilities.
- With exponential family likelihoods, the M-step only needs weighted sums of **sufficient statistics**.

Stochastic EM for the Gaussian mixture model

- On iteration i , grab a sub-sequence (aka **mini-batch**) of length M .
- E-step**: Compute the posterior probabilities for each data point in the mini-batch:

$$q(z_m = k) \leftarrow p(z_m = k \mid x_m, \Theta) \propto \frac{\pi_k \mathcal{N}(x_m \mid \mu_k, Q_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x_m \mid \mu_j, Q_j)}$$

Compute **normalized sufficient statistics** for the mini-batch:

$$\bar{T}_k^{(i)} = \frac{1}{M} \sum_{m=1}^M q(z_m = k) \quad \bar{\mathbf{t}}_{k,1}^{(i)} = \frac{1}{M} \sum_{m=1}^M q(z_m = k) x_m \quad \bar{\mathbf{t}}_{k,2}^{(i)} = \frac{1}{M} \sum_{m=1}^M q(z_m = k) x_m x_m^\top$$

Fold the normalized stats from this mini-batch into the running average via a **convex combination** with step size $\alpha \in [0,1]$:

$$\bar{T}_k \leftarrow (1 - \alpha) \bar{T}_k + \alpha \bar{T}_k^{(i)} \quad \bar{\mathbf{t}}_{k,1} \leftarrow (1 - \alpha) \bar{\mathbf{t}}_{k,1} + \alpha \bar{\mathbf{t}}_{k,1}^{(i)} \quad \bar{\mathbf{t}}_{k,2} \leftarrow (1 - \alpha) \bar{\mathbf{t}}_{k,2} + \alpha \bar{\mathbf{t}}_{k,2}^{(i)}$$

- M-step**: Update the parameters.

$$\mu_k \leftarrow \frac{\bar{\mathbf{t}}_{k,1}}{\bar{T}_k} \quad Q_k \leftarrow \frac{1}{\bar{T}_k} \left(\bar{\mathbf{t}}_{k,2} - \frac{\bar{\mathbf{t}}_{k,1} \bar{\mathbf{t}}_{k,1}^\top}{\bar{T}_k} \right)$$